

Supplementary Exam

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Electricity and Electronics EBN 111

July 2021

Test Information

Maximum marks:	46	Full marks:	45
Duration of test:	100 min	Open/Closed book:	Open
Total number of pages (this page included)		18	

Lecturers: Dr D. le Roux, Dr S. Smith, Prof X. Ye**Assistant Lecturers:** L. Kokot, M. Maritz, J. Thiselton, M. Trivedi, D. Wepener**Important**

1. The examination regulations of the University of Pretoria apply.
2. Write your answers on the Excel spreadsheet found on the AMS system. The spreadsheet contains unique parameter values for each student. Therefore, you must download the Excel spreadsheet from your own AMS account. Follow the instructions on the AMS spreadsheet carefully.
3. Final answers must be written as real numbers and not as fractions. Use at least 5 significant figures to write numbers, including complex numbers. Use decimal points and NOT decimal commas.
4. Answers must include units where applicable. (Refer to the EECE Rules for Electronic Assessments of tests and examinations for guidelines on how to enter units. Also refer to the example list of units on the Excel spreadsheet.)
5. **By uploading your semester test, you declare the following:**

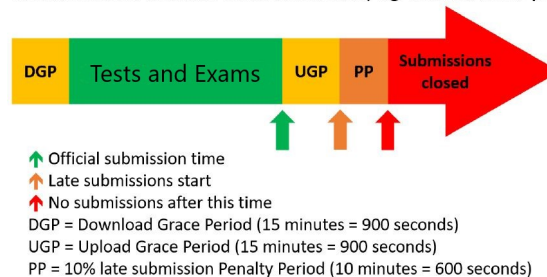
The University of Pretoria commits itself to produce academic work of integrity. I affirm that I am aware of and have read the Rules and Policies of the University, more specifically the Disciplinary Procedure and the Tests and Examinations Rules, which prohibit any unethical, dishonest or improper conduct during tests, examinations and/or any other forms of assessment. I am aware that no student or any other person may assist or attempt to assist another student, or obtain help, or attempt to obtain help from another student or any other person during tests, examinations and/or any other forms of assessment.

Online assessment submission rules and time-line

The following rules and time-line apply for all tests for EBN111/122 as illustrated by the figure below:

1. The Download Grace Period (DGP) is a 15 minute period before the official start of the test for students to download all documents related to the test and read all the necessary instructions.
2. The start of the green section after the DGP indicates the official start of the test. Students must work through the test at an average rate of at least 2 minutes per mark.
3. The upward green arrow at the end of the green section indicates the official submission deadline for the test.
4. Students must aim to submit their Excel spreadsheet with their answers on the AMS before the official submission deadline.
5. Students can request assistance via **e-mail** ebn2021.ams@gmail.com if they experience any technical issues with respect to uploading their Excel spreadsheets on the AMS.
6. An Upload Grace Period (UGP) of 15 minutes follows the official submission deadline. Students may submit their Excel spreadsheets with answers during this period without incurring any penalties. No email submissions will be accepted during the UGP.
7. A Penalty Period (PP) of 10 minutes follows the UGP. If a student submits an Excel spreadsheets during this period, the student incurs a 10% penalty to their final test mark.
8. No late submissions are accepted after the PP.

Late submissions of session-based assessments, e.g. tests and online practicals



Additional guide for Complex numbers

The following examples illustrate how to appropriately enter complex numbers on the Excel Spreadsheet:

- **Correct:** $1-j2$ or $-3+j4$
 - This is the rectangular form of a complex number. This format is acceptable as a single number entry in the numeric field.
 - Capital J is however incorrect, and $1-2j$ is also not acceptable.
 - If a complex number has no real part, for example $0+j2$, write $0+j2$ as the final answer.
- **Correct:**
 - With angle in degrees: $12\angle 34\deg$;
 - With angle in radians: $12\angle 0.5934$
 - A complex number in its polar form is regarded as a single number entry with the angle in degrees or radians.
- **Incorrect:**
 - $1-i2$ (Use j instead of i)
 - $1 - j2$ (no spaces between characters)
 - $12 \angle 34 \deg$; $34 \deg$; (no spaces between characters)
 - $j10+1$ (Real part before the imaginary part for rectangular form.)
 - $-3+4j$ (should be $j4$ instead of $4j$)

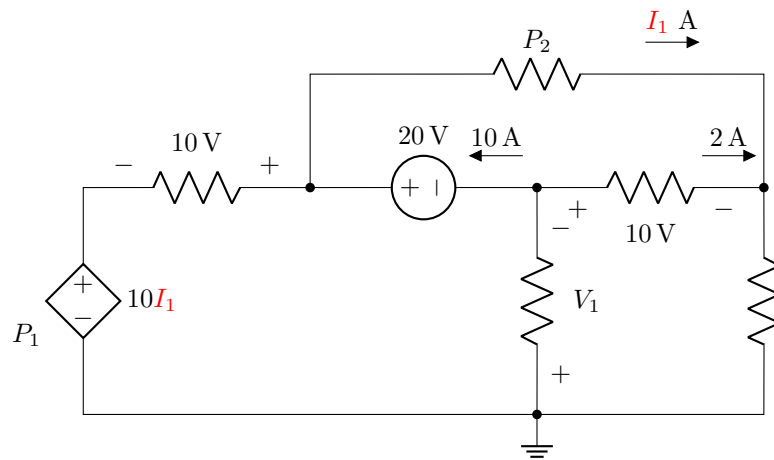
Section 1

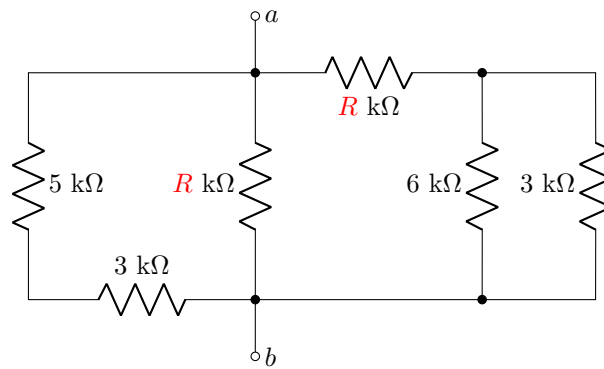
Question 1 [2]

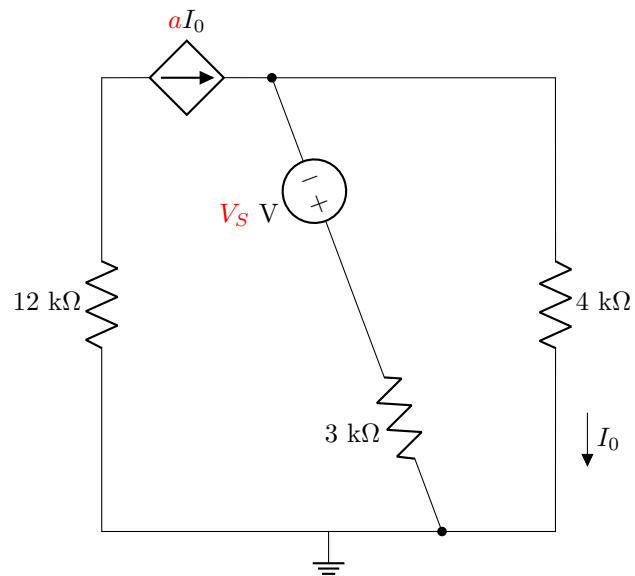
Given: x .

01	The charge entering the positive terminal of an element is $q(t) = 10 \sin(4\pi t)$ mC, while the voltage across the element is $v(t) = 2 \cos(4\pi t)$ V. What is the power associated with the element at $t = x$ s? [2]
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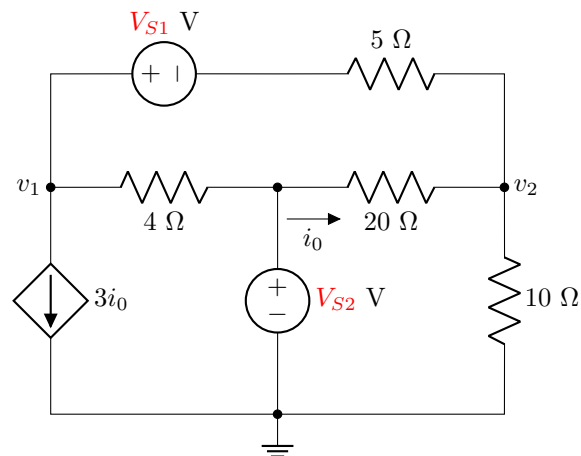
Questions 2 to 3 [4]

Given: I_1 .Find P_1 and P_2 .**02** P_1 [2]**03** P_2 [2]

Question 4 [2]Given: R .**04** Determine R_{ab} . [2]

Question 5 [2]Given: a , V_S .**05** Use either nodal or mesh analysis to determine I_0 . [2]

Questions 6 to 9 [4]

Given: V_{S1} , V_{S2} .

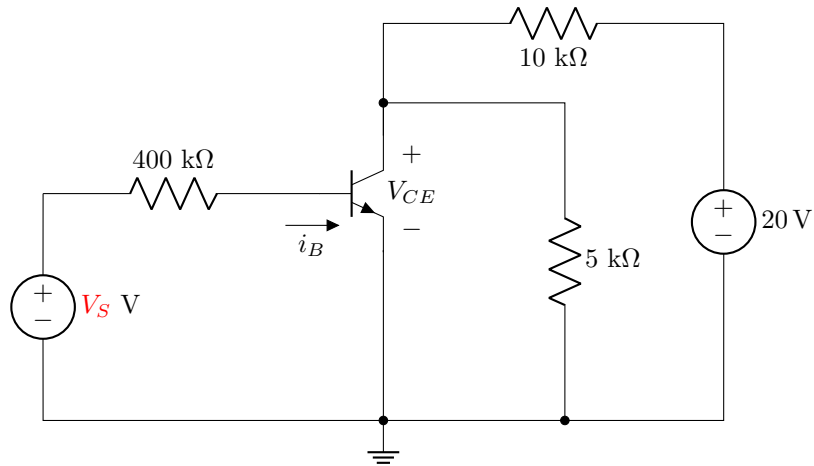
Use nodal analysis to set up the following two equations of the form:

$$9v_1 + av_2 = b$$

$$cv_1 + 7v_2 = d$$

Calculate the unknown coefficients a , b , c and d .

06	a [1]	07	b [1]	08	c [1]	09	d [1]
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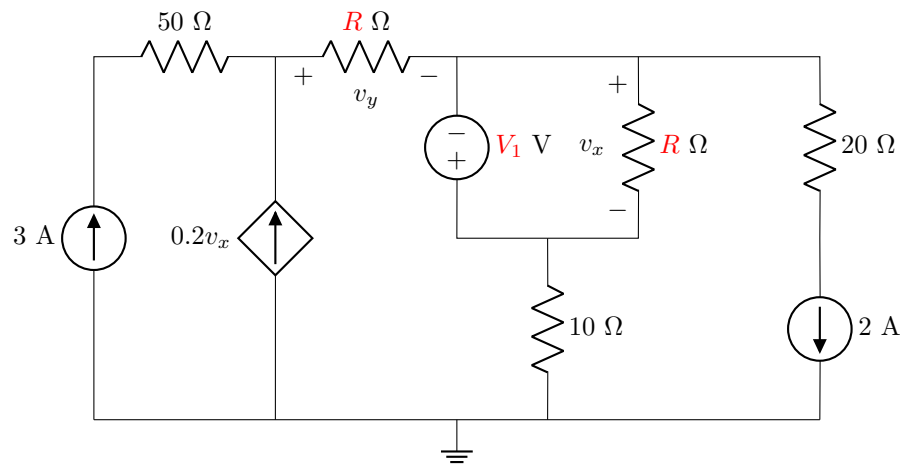
Question 10 [2]Given: V_S It is known that $\beta = 100$ and $V_{BE} = 0.7$ V.**10** Determine V_{CE} . [2]**Total Section 1: [16]**

Section 2

Question 11 [2]

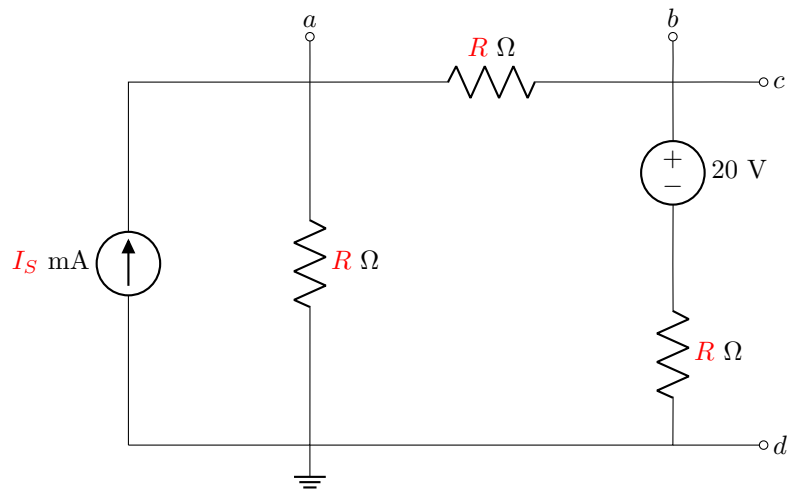
Given: V_1 , R .

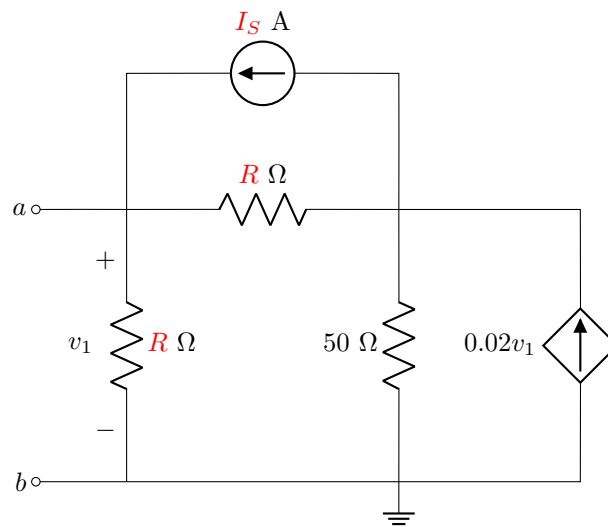
Determine $v_y(V_1)$, the independent contribution of V_1 to v_y .

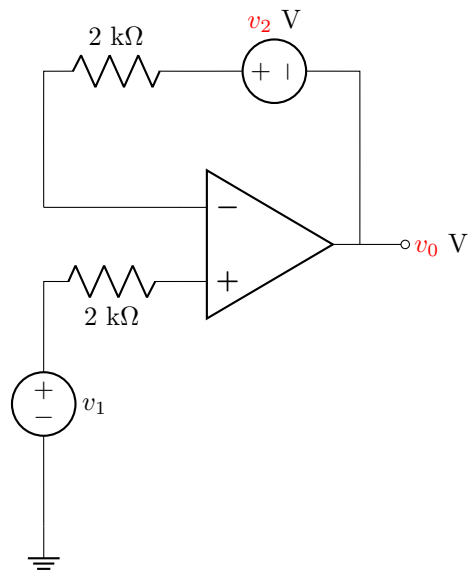


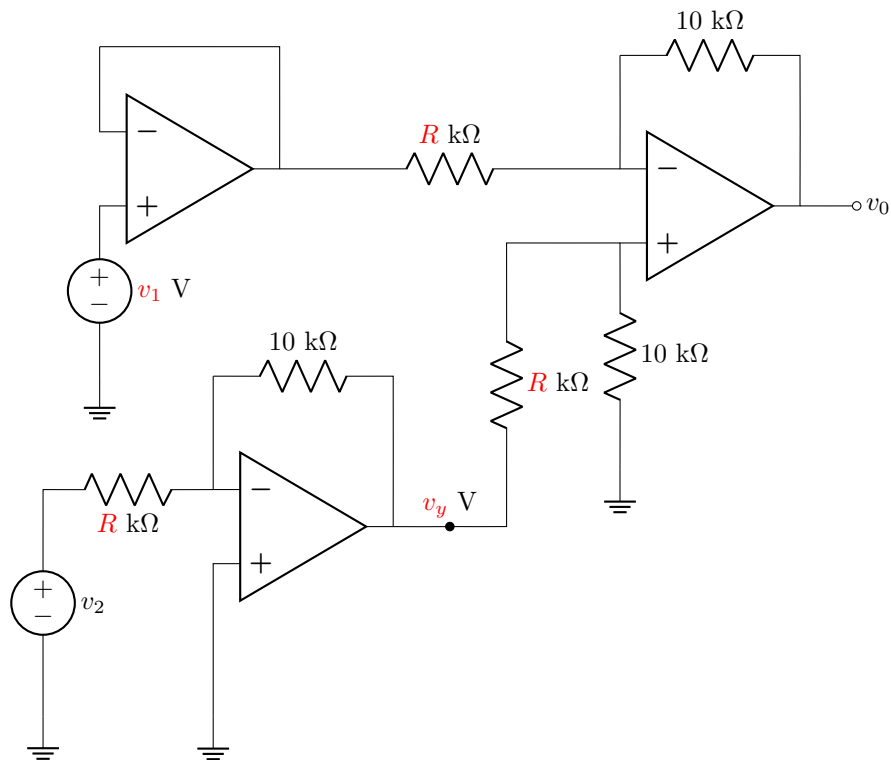
11	$v_y(V_1)$ [2]
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Questions 12 to 13 [4]

Given: I_S , R .Determine the Thevenin equivalent voltage V_{Th} and resistance R_{Th} between terminals c - d .12 V_{Th} [2]13 R_{Th} [2]

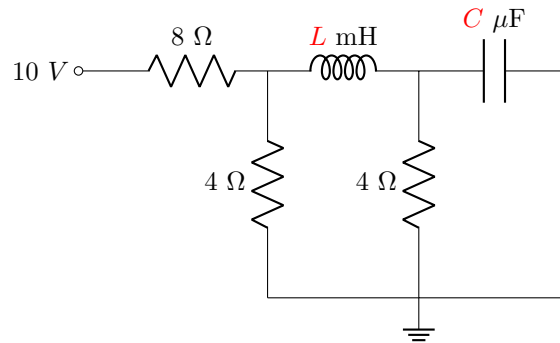
Question 14 [2]Given: I_S , R .Determine the Norton equivalent current I_N between terminals a - b .**14** I_N [2]

Question 15 [2]Given: v_2 , v_0 .Assume that the op-amp is ideal. Determine v_1 .**15** | v_1 [2]

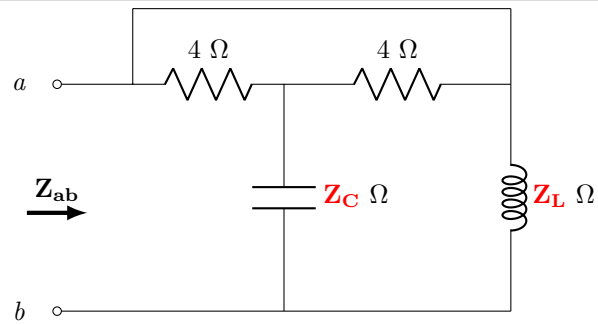
Questions 16 to 17 [4]Given: R , v_1 , v_y .Assume that the op-amps are ideal. Calculate the voltages v_2 and v_0 .**16** v_2 [2]**17** v_0 [2]**Total Section 2: [14]**

Section 3**Questions 18 to 19 [4]**Given: L , C

The circuit below operates under steady dc conditions. Find the energy w_L stored in the inductor, and the energy w_C stored in the capacitor.

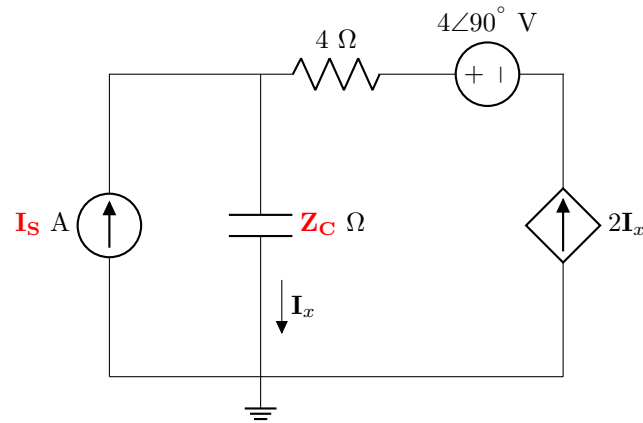
**18** w_L [2]**19** w_C [2]

Questions 20 to 21 [4]	
Given: A , θ	
Consider the following two sinusoids: $v_1(t) = -A \sin(\omega t + 60^\circ) \text{ V, and}$ $v_2(t) = 10 \cos(\omega t - \theta^\circ) \text{ V, } \theta^\circ \in [0^\circ, 25^\circ]$ The two sinusoids are represented by phasors $\mathbf{V}_1$ and $\mathbf{V}_2$, respectively. It is also known that $v_3(t)$ leads $v_2(t)$ by 50°.	
20	Choose one correct statement from the following list. 1. v_1 leads v_2 by $(150+\theta)^\circ$. 2. v_2 leads v_1 by $(150-\theta)^\circ$. 3. v_1 leads v_3 by $(50+\theta)^\circ$. 4. v_3 leads v_1 by $(50-\theta)^\circ$. Write i as your answer if statement i is correct, i=1, 2, 3, and 4. [2]
21	Use phasors to find $\mathbf{V} = \mathbf{V}_1 + \mathbf{V}_2$ in polar form . [2]

Question 22 [2]Given: $\mathbf{Z_L}$, $\mathbf{Z_C}$.**22** Determine $\mathbf{Z_{ab}}$ in rectangular form. [2]

Question 23 [2]Given: $\mathbf{I_S}$, $\mathbf{Z_C}$.

Find $\mathbf{I}_x^A$, the contribution to $\mathbf{I}_x$ from the **independent** current source. Write your answer in **rectangular form**.



23 | $\mathbf{I}_x^A$ (rectangular form). [2]

Questions 24 to 25 [4]	
Given: $\mathbf{Z_L}$, $\mathbf{Z_C}$	
Find the Norton equivalent current $\mathbf{I_N}$ and impedance $\mathbf{Z_N}$ at terminals a - b . Write your answers in rectangular form.	
24	$\mathbf{I_N}$ (rectangular form) [2]
25	$\mathbf{Z_N}$ (rectangular form) [2]

Total Section 3: [16]

Grand Total : [46]